

ATPL EASA STUDY GUIDE | 040 - HUMAN PERFORMANCE

Chapter 7: Toxic Hazards & Pilot Health

Syllabus Source: CAE Oxford (pp. 205-236) | Pass Target: >= 90% AviationExam

1. CARBON MONOXIDE (CO) POISONING IN AVIATION

Carbon Monoxide is a colorless, odorless, tasteless, non-irritating toxic gas produced by incomplete hydrocarbon combustion. In light aircraft, exhaust gas shroud heaters are the most common source of in-flight CO leaks:

- Hemoglobin Affinity: CO binds to hemoglobin with an affinity 200 to 250 TIMES GREATER than oxygen, forming Carboxyhemoglobin (COHb).
- Mechanism of Hypoxia: CO produces both ANEMIC HYPOXIA (reduces O2 capacity) and HISTOTOXIC HYPOXIA (blocks cellular cytochromes). Additionally, it shifts the O2 dissociation curve to the LEFT, locking remaining oxygen onto hemoglobin!
- Clinical Symptoms: Initial: Dull frontal headache, sluggishness, loss of concentration. Intermediate: Throbbing temporal headache, dizziness, nausea, impaired motor coordination. Late: Confusion, cherry-red skin/lips (usually post-mortem), convulsions, coma, death.
- In-Flight Treatment: 1) Turn cabin heat OFF immediately; 2) Open 100% fresh air vents; 3) Don oxygen mask with 100% O2 under positive pressure; 4) Land at nearest suitable aerodrome.

2. ALCOHOL (ETHANOL) & AEROMEDICAL REGULATIONS

Alcohol is a powerful central nervous system (CNS) depressant. It is rapidly absorbed in the stomach and small intestine, and metabolized in the liver by alcohol dehydrogenase at a CONSTANT RATE of ~10 to 15 mg% (0.010% to 0.015%) per hour.

[!] EXAM TRAP (AVIATIONEXAM) - EASA Blood Alcohol Limits & The 'Bottle-to-Throttle' Rule

Under EASA Part-CAT and Part-MED regulations: - Maximum Legal Blood Alcohol Concentration (BAC): 0.20 g/l (0.02% or 20 mg/100 ml). Absolute zero is strongly recommended. - Bottle-to-Throttle Rule: Minimum 8 HOURS between the consumption of any alcohol and reporting for flight duty. - Post-Alcohol Effects (Hangover): A pilot is UNFIT to fly even when BAC reaches 0.00% if fatigue, dehydration, or headache persists. Alcohol alters endolymph density in the inner ear for up to 24-48 hours, inducing Positional Alcohol Nystagmus (PAN) and extreme vertigo!

3. MEDICATIONS & FLYING RESTRICTIONS

Medication Category	Common Usage	Aeromedical Flying Privilege Status
Antihistamines	Allergies, motion sickness	STRICTLY DISQUALIFYING. Induce profound sedation, drowsiness, and degraded cognitive reaction times.
Antibiotics	Bacterial infections	PILOT GROUNDED. Disqualification is based both on the drug's side effects (nausea, diarrhea) and the underlying infectious illness.
Analgesics (Painkillers)	Headache, muscle ache	Mild analgesics (Paracetamol, Aspirin) are acceptable in small doses. Strong analgesics containing CODEINE, opioids, or sedatives are PROHIBITED.

Decongestants	Nasal congestion	Oral/nasal pseudoephedrine may cause palpitations, nervousness, and rebound congestion; masking ear block risks.
Antidepressants & Sedatives	Depression, insomnia	Benzodiazepines, sleeping pills, and unapproved SSRIs are STRICTLY DISQUALIFYING.

4. FOOD HYGIENE & CREW MEAL SEPARATION

Gastrointestinal illness and food poisoning (e.g. Salmonella, Staphylococcal enterotoxin) represent a severe hazard for single-pilot incapacitation. In commercial multi-pilot airline operations, airline Standard Operating Procedures (SOPs) strictly mandate that the Captain and First Officer MUST eat DIFFERENT meals prepared from different kitchens or food sources, served at different times before and during flight.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Algoritmo Paso a Paso (Guía para Dummies): SHELL Model & Error Analysis

REGLA MNEMOTÉCNICA: L-H-S-E-L Paso 1: Identificar siempre el Liveware Central (L) como el piloto/operador humano. Paso 2: Si el problema involucra mandos, pantallas o botones confusos -> Es un desajuste L-H (Hardware). Paso 3: Si involucra una checklist mal redactada o un manual confuso -> Es un desajuste L-S (Software). Paso 4: Si involucra turbulencia, calor o hipoxia en cabina -> Es un desajuste L-E (Environment). Paso 5: Si involucra mala coordinación entre Capitán y Primer Oficial o con ATC -> Es un desajuste L-L (Liveware).

5. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: Why is Carbon Monoxide (CO) particularly dangerous in aviation?

- [A]: It has an irritating smell that causes coughing
- [B]: It is odorless, colorless, and binds to hemoglobin with ~250 times greater affinity than oxygen
- [C]: It causes immediate lung rupture
- [D]: It only affects flights above FL 400

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. CO is insidious (odorless, tasteless) and binds strongly to hemoglobin, depriving tissues of oxygen (anemic and histotoxic hypoxia).

Q2: What is the maximum permitted blood alcohol concentration (BAC) for flight crew under EASA regulations?

- [A]: 0.50 g/l (0.05%)
- [B]: 0.80 g/l (0.08%)
- [C]: 0.20 g/l (0.02%)
- [D]: 0.00 g/l with no tolerance

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [C]. EASA Part-MED / Part-CAT specifies a legal upper limit of 0.20 g/l (0.02%), although zero tolerance is operational policy.

Q3: What is the minimum recommended 'bottle-to-throttle' time interval between drinking alcohol and flight du

- [A]: 4 hours
- [B]: 8 hours
- [C]: 12 hours
- [D]: 24 hours

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. 8 hours is the mandatory minimum time interval between alcohol consumption and flight duty.

Q4: Can the metabolic rate of alcohol elimination in the human liver be accelerated by drinking black coffee or

- [A]: Yes, coffee speeds up liver enzymes by 50%
- [B]: Yes, heavy sweating eliminates alcohol through pores
- [C]: NO, alcohol metabolism proceeds at a fixed rate of ~10-15 mg/100 ml per hour and cannot be accelerated
- [D]: Yes, with pure oxygen therapy

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [C]. Hepatic oxidation of ethanol by alcohol dehydrogenase is zero-order (constant rate) and cannot be accelerated.

Q5: What is the primary aeromedical hazard of taking over-the-counter antihistamines before flying?

- [A]: High risk of blood clotting
- [B]: Sedation, drowsiness, and impaired psychomotor reaction time
- [C]: Eye pupil constriction
- [D]: Hypertension

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. Classic antihistamines cross the blood-brain barrier and cause marked sedation, slowing reaction times and inducing sleepiness.

Q6: Why must the Pilot-in-Command and the Co-pilot consume different meals on a commercial flight?

- [A]: Airline budget optimization
- [B]: To prevent simultaneous food poisoning incapacitating both pilots
- [C]: Personal preference of senior captain
- [D]: Union dietary contracts

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. Consuming different meals prepared from different sources prevents dual crew incapacitation from food contamination.